	Target engagement	Dose	Efficacy	Safety
#15 **gastric cancer** anti-ERBB2 antibodies (3)	**2017** Anti-HER2 bispecific: cell proliferation in vitro and xenografts Not measured in patients	**2018** Cynomolgus single and repeat-dose tox (margetuximab)	**2025** Anti-HER2 bispecific: cell proliferation in vitro and xenografts △ JACOB missed overall survival	**2018** Cynomolgus single and repeat-dose tox (margetuximab)
#16 **hepatitis C** HCV NS3/4A protease inhibitors (13)	**2012** Viral-kinetic PK/PD model driven by replicon EC50, plasma… Decline under-predicted	**2009** Replicon EC99 vs rat plasm… Once-daily target reached, tolerated, active	**2016** Human hepatocyte chimeric mouse (uPA/SCID) infected with patient HCV genotype 1b	**2016** Rodent and dog repeat-dose toxicology
hepatitis C HCV NS5A inhibitors (10)	**2010** HCV replicon NS5A resistance phenotyping (Huh-7 replicons carrying replicon-selecte…	**2015** HCV subgenomic replicon (Huh-7 cells; genotype 1a/1b)	**2020** HCV viral-kinetic model of on-treatment HCV RNA decline, later run in real time to set DAA duration	**2014** Rodent, dog and monkey tox △ Rat/monkey adrenal signal abse…
hepatitis C NS5B non-nucleoside inhibitors (10)	**2014** HCV genotype 1 subgenomi… Dasabuvir monotherapy cut HCV RNA 1.4 log10	**2014** Rat and dog repeat-dose… 10-120x animal margin held in patients	**2012** HCV genotype 1 subgenomic replicon (Huh-7), with and without… △ Replicon potent; filibuvir no SVR	**2014** Rat and dog repeat-dose… No animal target organ, none in patients
hepatitis C HCV NS5B (nucleotide) inhibitors (7)	**2012** HCV subgenomic replicon… HCV RNA fell dose-dependently in 3 days	**2013** Standard repeat-dose toxicology and safety pharmacology (mouse, rat, dog, cynomolgus…	**2013** HCV subgenomic replicon (Huh-7 clone A; genotype 1b and chimeric 1-6 replicons)	**2025** Recombinant human CYP3A4 metabolite-formation assay… △ Tox missed BMS-986094 heart…
hepatitis C IFNAR1/IFNAR2 agonists (3)	**2011** Rhesus/cynomolgus monkey PK and serum antiviral activity	**2014** Rhesus monkey and rat PK No checkable prediction	**2011** Rhesus/cynomolgus monkey PK and serum antiviral activity	**2011** PEG-IFN-SA nonclinical toxicology Model data unpublished
#17 **diabetic kidney disease** NR3C2 antagonists (3)	**2021** Rat urinary Na+/K+ natriuresis under MR-agonist challenge	**2021** Phase 2b population PK/exposure-response model of UACR, serum potassium and eGFR	**2020** Uninephrectomised DOCA-salt rat △ Post-FIDELIO ZSF1 rat misses…	**2021** Rat autoradiography and equinatriuretic DOCA-salt △ No rat hyperkalaemia; DKD had it
#18 **acute myeloid leukaemia** FLT3 inhibitors (4)	**2013** FLT3-ITD cell lines and MV4-11 xenografts Patient plasma fully blocks FLT3-ITD	**2013** FLT3-ITD cell lines and MV4-11 xenografts	**2020** FLT3-ITD cell lines and… Gilteritinib + azacitidine: no survival gain	**2018** Rat and dog toxicology and safety pharmacology PRES and QT missed by tox species
acute myeloid leukaemia CD33-directed depleters (3)	**2021** AML blasts + NK cells ex vivo NK activation unchanged despite CD33 saturation	**2017** CD33+ AML xenografts (mice) No checkable prediction	**2021** Cord-blood CD33-CAR-NK: Molm13 xenografts, HSC colonies	**2018** CD33+ AML xenografts (mice)
#19 **cervical cancer** anti-PDCD1 antibodies (4)	**2021** Cynomolgus PD-1 receptor… Human PD-1 occupancy >80%, as in monkeys	**2024** BIOCAD population PK model simulating a 250 mg Q3W flat dose of prolgolimab	**2024** CT26 colon tumours in mice (prolgolimab vs pembrolizumab) CAESURA ORR 63.8%, combination	**2022** Cynomolgus 26-week repeat-dose toxicology… Tolerated, as in cervical patients
#20 **malignant pancreatic neoplasm** MAP2K1/2 inhibitors (3)	**2012** Pancreatic and other tumour xenografts (cell-line, orthotopic and patient-derived)	**2012** Pancreatic and other tumour xenografts (cell-line, orthotopic and patient-derived)	**2014** Pancreatic and other tumour xenografts (cell-line, orthotopic an… △ PDX regressions; no patient…	**2013** Rat, dog, rabbit, mouse safety △ Unpredicted retinal harm with…
#21 **MASH** NR1H4 agonists (4)	**2017** Rodent FXR target-gene P… Volunteer FGF19/C4 reproduced (retrodiction)	**2019** STAM and AMLN-diet NASH mice (tropifexor) No checkable prediction	**2019** Choline-deficient high-fat diet plus nitrite NASH rats (cilofexor) Rat fibrosis fell; ATLAS 12% vs 11%	**2021** Humanized chimeric-liver (PXB) mice, plasma lipoproteins △ Cilofexor LDL rise contradicts H…
MASH FGF21 analogues (3)	**2016** db/db and diet-induced obese mice	**2023** Pegozafermin population PK/PD exposure-response model	**2023** Pegozafermin population PK/PD exposure-response model △ Pegbelfermin failed on fibrosis	**2020** FGF21 variant carriers (UK Biobank) △ No mouse bone loss at 16 weeks
#22 **prostate cancer** AR antagonists (5)	**2012** LNCaP/AR castrate xenografts PSA fell >=50% in 46.7%	**2012** LNCaP/AR castrate xenografts	**2013** ARN-509/enzalutamide-resistant cell lines and AR F876L xenografts	**2019** Mouse brain:plasma distribution with rat and dog CNS toxicology (darolutamide)
prostate cancer PARP1/2 inhibitors (5)	**2009** BRCA2-null V-C8, ES cells, xenografts; PARP inhibitor alone	**2015** BRCA2-null V-C8, ES cells, xenografts; PARP inhibitor alone	**2022** BRCA2-null V-C8, ES cells, xenografts; PARP inhibitor alone △ Non-BRCA HRR genes: rPFS H…	**2019** Rat, dog repeat-dose tox of olaparib, niraparib, talazoparib △ Pamiparib prostate: anaemia 7/1…
prostate cancer CYP17A1/AR inhibitors (3)	**2016** Recombinant CYP17A1 17alpha-hydroxylase-vs-17,20-lyase enzyme assay rating VT-464…	**2018** VCaP xenografts in mice Exposure plateaued; PK stopped development	**2016** Circulating-tumour-cell… AR-V7-selected phase 3 stopped for futility	**2018** Cynomolgus monkey and hCG-treated rat adrenal/testicular… △ CNS tox closed prostate phase 2
prostate cancer EZH2 inhibitors (3)	**2019** CRPC lines and xenografts, EZH2i + AR blockade	**2012** CRPC lines and xenografts, EZH2i + AR blockade No checkable prediction	**2024** CRPC lines and xenografts,… Tazemetostat + enzalutamide missed rPFS	**2021** Rat and monkey repeat-dose toxicology (tazemetostat)
prostate cancer FOLH1-directed depleters (3)	**2015** PSMA-positive vs PSMA-negative prostate tumour-bearing mice (LNCaP/PC3…	**2011** PSMA-high CRPC xenografts, PSMA ADC No checkable prediction	**2024** PSMA-positive vs… ARROW: PSA50 63% vs 31% with enzalutamide	**2022** GLP single-dose rat, minipig and 4-week rat tox, cold PSMA-617 △ Ex-vivo gland signal minimal; P…
#23 **follicular lymphoma** MS4A1-directed depleters (4)	**2019** Cynomolgus monkey B-cell depletion / toxicology B cells depleted in patients	**2019** Cynomolgus monkey B-cell depletion / toxicology	**2022** Semimechanistic PBPK/PD trimer model (Genmab)	**2019** Cynomolgus monkey B-cell depletion / toxicology
follicular lymphoma CD19-directed depleters (3)	**2010** Syngeneic BALB/c A20 lymphoma, 6 Gy TBI, first-generation anti-mouse-CD19 CAR T cells	**2015** Raji xenograft in NSG mice, CD19 CAR T No checkable prediction	**2024** Raji xenograft in NSG mice, CD19 CAR T	**2021** SCID-beige mice with high Raji burden, 19-28z CAR T cells… △ CRS/neurotox despite lenzilumab
follicular lymphoma PIK3CD inhibitors (3)	**2011** Patient B-cell tumour samples and B-cell lines ex vivo	**2024** Healthy-volunteer whole-blood basophil activation test, ex-vivo PI3Kδ PD assay for ME-401	**2013** Patient B-cell tumour samples and B-cell lines ex vivo	**2013** p110δ kinase-dead (D910A) knock-in mouse △ Heart, pancreas harms predicte…
#24 **ovarian cancer** PARP1/2 inhibitors (6)	**2009** Human tumour xenografts, needle-biopsy PAR ELISA (veliparib) PAR fell in patient tumour biopsies	**2021** NOVA exposure–safety model (niraparib)	**2016** BRCA-deficient cancer cells… ARIEL2: BRCA-mutant PFS 12.8 months	**2016** BRCA1-deficient xenograft… Marrow toxicity contradicted 'well tolerated'
#25 **renal cell carcinoma** VEGFR multikinase inhibitors (4)	**2012** Rat GS9L glioma DCE-MRI… Patient tumour perfusion fell on DCE-MRI	**2010** Human xenografts in nude mice (786-O renal and others) No checkable prediction	**2024** Lenvatinib 19-model human xenograft panel with vascular scor… Not measured in patients	**2015** Rat and monkey repeat-dose toxicology (lenvatinib) △ Not well tolerated in RCC patients
#26 **rheumatoid arthritis** BTK inhibitors (8)	**2013** BTK occupancy probe in CI… Full BTK occupancy in humans, as in mice	**2022** CIA occupancy-efficacy PK/PD No checkable prediction	**2019** Collagen-induced arthritis (rodent) △ Missed ACR20 at week 12	**2021** Human platelet aggregometry and humanized-mouse thrombus… Not measured in patients
rheumatoid arthritis SYK inhibitors (3)	**2018** Cynomolgus monkey platelet/bleeding-time (lanraplenib) Not measured in patients	**2018** Rat collagen-induced arthritis, therapeutic dosing from peak disease (lanraplenib/GS-9876)	**2018** Rat collagen-induced arthritis, therapeutic dosing from peak… DAS28-CRP no better than placebo	**2018** Cynomolgus monkey platelet/bleeding-time (lanraplenib) △ Serious infections missed, ended
rheumatoid arthritis anti-CSF2 antibodies (3)	**2018** Mouse GM-CSF-IRF4-CCL17 pathway in zymosan/antigen arthritis pain models	**2025** Cynomolgus 13-week SC and 26-week IV GLP tox, otilimab △ Q2W regimen lost engagement	**2018** Mouse arthritis, anti-GM-CSF △ Failed in targeted-therapy-IR; m…	**2019** Human anti-GM-CSF autoantibody alveolar proteinosis △ No PAP, 2915 patients, 2.5 years
#27 **multiple myeloma** TNFRSF17-directed depleters (8)	**2015** BCMA+ myeloma xenografts in NSG mice, CAR-T	**2026** Belantamab PopPK, tumour and keratopathy model No checkable prediction	**2019** BCMA+ myeloma xenografts in NSG mice, CAR-T	**2020** Rat and rabbit repeat-dose toxicology (belantamab mafodotin)
multiple myeloma PSMB5 inhibitors (3)	**2012** Human MM.1S myeloma… Whole-blood 20S inhibition seen in patients	**2015** Ixazomib population PK, fixed 4 mg versus BSA-based dosing	**2014** Patient-derived primary CD138+ myeloma cells ex vivo,… Not measured in patients	**2017** Delanzomib programme:… Marrow sparing predicted; thrombocytopenia DLT
#28 **gout** SLC22A12 inhibitors (5)	**2019** Tufted capuchin monkeys (low uricase, high plasma urate)… Volunteers: urate down, FEUA up	**2015** Rat and cynomolgus monke… Renal failure capped dose inside margin	**2013** Tufted capuchin monkeys (low uricase, high plasma urate) dosed with UR-1102/epaminurad	**2025** In vitro hepatotoxicity-mechanism assays (mitochondrial toxicity, reactive…
#29 **melanoma** BRAF inhibitors (3)	**2010** BRAF V600E cell lines and xenografts	**2011** BRAF V600E cell lines and xenografts	**2009** BRAF V600E vs BRAF-wild-type cell-line panel… △ INTERIM: intermittent worse again	**2014** RAS-mutant cells, keratinocytes and mouse skin… △ Rat skin toxicity matched; cuSC…
melanoma anti-PDCD1 antibodies (3)	**2020** Humanised-PD-1 C57BL/6 mouse with S180 homografts, 124I-toripalimab immuno-PET	**2019** Human T-cell PD-1 occupancy in vitro (saturation threshold)	**2024** Human PD-1 knock-in BAL… Post-hoc mouse reproduces melanoma activity	**2024** Cynomolgus monkey (PD-1… Endocrine and skin irAEs in melanoma cohort
#30 **Crohn disease** anti-IL23A antibodies (3)	**2018** T-cell transfer colitis mouse (anti-mouse IL-23p19 mAb)	**2022** Sponsor population-PK / exposure-response dose model from phase 2, risankizumab, mirikizumab	**2017** T-cell transfer colitis mouse (anti-mouse IL-23p19 mAb)	**2022** Monkey tox, risankizumab Hepatotoxicity in Crohn's; monkeys silent
#31 **multiple sclerosis** MS4A1-directed depleters (4)	**2009** Cynomolgus monkey B-cell depletion PK/PD and toxicology	**2022** Population PK-B cell model of subcutaneous ofatumumab in RMS (pooled phase 2/3, 1486 patients)	**2025** Population exposure-response analysis of… Higher dose gave no added benefit	**2009** Cynomolgus monkey B-cell depletion PK/PD and toxicology